

 **1st RANK HACKATHON TEMPLATE**

PROBLEM: SIH26068

DISASTER MANAGEMENT & AI

WeatherGPT

Conversational AI for Weather Forecasting, IMD Alerts & Climate Information

Team Credentials

Team Name: Team WeatherGPT

Lead Developer: Dev Vrajawat

Domain: AI & Climate Resilience

Innovation Focus

Chat-First Natural Language Assistant, 700+ District Coverage, IMD Severe Weather Alert Engine & Voice Multimodal UI.

Project Deliverables

Live Web App:

devvrajawat0.github.io/weathergpt

GitHub Repo:

github.com/devvrajawat0/weather-gpt-2

The Problem Context

Traditional weather applications present fragmented numeric charts, lack conversational context, incur heavy geocoding API costs, and fail to provide accessible voice advisories or severe disaster alerts for non-technical or rural populations.

Our Proposed Solution: WeatherGPT

A chat-first climate intelligence platform allowing users to ask complex questions ("Will it rain in Bhopal tomorrow?", "Compare weather in Delhi and Tokyo") and receive conversational answers alongside visual dashboards.

Complete Data Coverage

- **700+ Indian Districts:** Grouped by State/UT with exact coordinates.
- **195+ World Capitals:** Complete global capital city coverage.
- **Live Feeds:** Temp, Feels-like, Wind, Humidity, UV Index, AQI, Sunrise/Sunset.

Prototype UI Showcase

Live App Screenshots

{/* MOCK PROTOTYPE CARD DISPLAY */}

IN Bhopal, Madhya Pradesh

24.5°C • Light Drizzle

24.5°C

Feels like 25.1°C | Humidity 88%



YELLOW ALERT: Moderate Rainfall Advisory

Precipitation (25.9mm) forecast. Carry rain gear and drive with caution.

Air Quality Index (AQI): 48 (Good)

PM2.5: 12 µg/m³

PM10: 25 µg/m³

O3: 30 µg/m³

⚙️ Production Tech Stack

React 18

TypeScript

Vite

Tailwind CSS

Node.js

Express

TanStack Query

Leaflet Maps

Anthropic SDK

Open-Meteo API

- **Frontend Layer:** Component-driven React UI with dynamic weather gradients, mobile-first breakpoints, dark mode, and Leaflet map rendering.
- **Backend API Proxy:** Node.js Express server with rate limiting (`express-rate-limit`), CORS security, and 15-minute response caching (`node-cache`).
- **AI & Fallback Layer:** Anthropic Claude API proxy with real-time weather context injection + built-in Intelligent Fallback AI Engine.

🗣️ Web Speech API & Multimodal Input

- **Voice Input (Speech-to-Text):** Native microphone voice search integration.
- **Text-to-Speech (TTS):** Reads natural-language weather advisories aloud.

🤖 WeatherGPT Conversational AI Interface

Will it rain in Bhopal tomorrow?

☁️ **Rain Forecast for Bhopal, MP:**

- **Today:** 100% rain chance (Max: 25.1°C)

- **Tomorrow:** 92% rain chance (Max: 28.1°C)

☔ **Advice:** Umbrella recommended. Drive carefully.

🔄 Data Pipeline Flowchart



💰 Zero-Cost Geocoding Innovation

Commercial geocoding APIs (e.g. Google Maps API) incur massive operational fees and strict rate limits. WeatherGPT solves this by pre-bundling latitude/longitude coordinate matrices for **700+ Indian Districts** and **195+ World Capitals** directly into JavaScript modules, delivering **zero API cost** and **zero lookup latency**.

⚡ Double-Layer Caching & Quota Efficiency

- ➔ **Backend Server Cache:** `node-cache` (15-min TTL) cuts external API calls by >90%.
- ➔ **Frontend Client Cache:** TanStack React Query maintains an in-memory stale time.
- ➔ **Free Open-Meteo API Tier:** 10,000 requests/day per IP ensuring extreme financial viability.

🛡️ 100% Offline & Static Resilience

If the Express backend server is offline (e.g. static hosting on GitHub Pages or Surge), WeatherGPT automatically falls back to direct browser Open-Meteo calls and client-side AI response synthesis with **zero downtime**.

📊 ROI & Performance Comparison Matrix

Metric	Traditional Weather Apps	WeatherGPT Solution
Geocoding Cost	High Paid API Fees	₹0 (Pre-Bundled Static Dataset)
Search Latency	1,500ms – 3,000ms	<10ms (Instant Local Lookup)
Voice Input/TTS	Not Supported	Web Speech API Speech & Audio TTS
Offline Resilience	Fails completely	100% Client-Side Fallback

Agricultural Weather Support for Farmers

Provides tailored farming advisories covering irrigation scheduling, wind-drift pesticide warnings, and crop harvest shelter guidance ahead of predicted rain showers.

Disaster Management & Emergency Preparedness

Delivers early warnings for extreme heatwaves, flash floods, squalls, and AQI spikes with actionable safety instructions and disaster helpline contacts (NDRF: 1078, IMD: 1800-180-1717).

Accessibility & Digital Inclusion

Hands-free voice input and text-to-speech reading allow rural populations and non-literate users to access critical weather advisories effortlessly.

Real Prototype Navigation

4 Live Views

Dashboard

Current weather, 24h ribbon, 7-day bars, Leaflet map

Chat AI

Voice-in, speech TTS, natural language Q&A

Explorer

700+ Indian districts & 195+ world capitals

IMD Alerts

Color-coded Red/Orange/Yellow warnings

Live Prototype URL: devvrajawat0.github.io/weathergpt

Standards & Scientific Frameworks

- **IMD Alert Criteria:** India Meteorological Department warning thresholds for Heatwaves ($\geq 40^{\circ}\text{C}$), Heavy Rainfall ($\geq 64.5\text{mm}$), and Squalls.
- **CPCB Air Quality Index:** Central Pollution Control Board standards for US AQI, PM2.5, PM10, and gaseous pollutants.
- **WMO Weather Codes:** World Meteorological Organization code mapping for international weather condition standardization.

Meteorological API References

- Open-Meteo Weather Forecast API: `open-meteo.com/en/docs`
- Open-Meteo Air Quality API: `open-meteo.com/en/docs/air-quality-api`

Deliverables & Live Links Matrix

Deliverable	URL / Location
Live Web App	<code>devvrajawat0.github.io/weathergpt/</code>
GitHub Repository	<code>github.com/devvrajawat0/weather-gpt-2</code>
Surge Deployment	<code>weathergpt2.surge.sh</code>

SIH26068 • Think • Build • Innovate • Impact IN